

**ENSC 225 – Lab #1**

**Operational Amplifiers**

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## Abstract

This lab report presents the findings from ENSC 225 Laboratory #1, which focuses on the characterization and application of operational amplifiers (op-amps). Utilizing a  $\pm 12$  Volt supply and adhering to a current limit of 80 mA, the experiment explores various configurations of op-amp circuits using the TL084 chip. The primary objectives include designing non-inverting and inverting amplifiers, measuring key parameters such as input offset voltage, bias current, and offset current, and implementing circuits like the Schmitt trigger and free-running oscillator. Prior to experimentation, proper calibration of the oscilloscope and function generator is essential to ensure accurate measurements.

In the initial phase of the lab, participants will construct amplifiers with specified gains and analyze their performance by comparing experimental results to theoretical predictions based on measured resistor values. The lab also includes a frequency response analysis to determine how the op-amp behaves across a range of frequencies, as well as an investigation into DC imperfections using a multi-op-amp chip. Finally, the construction and simulation of both a Schmitt trigger and a free-running oscillator will allow for practical applications of the op-amps studied, providing valuable insights into their operational characteristics in real-world scenarios.

## Experiment 1

**Introduction:** In this experiment, both non-inverting and inverting amplifiers will be constructed with gains of approximately  $|10|$  and  $|100|$ , using resistors between  $1\text{K}\Omega$  and  $100\text{K}\Omega$ . A 1 kHz sinusoidal wave will be applied to the amplifiers using a function generator, and the input and output voltages will be observed via an oscilloscope.

The objective is to compare the experimental gain values to the theoretical values calculated from the actual measured resistor values, assessing the accuracy of the amplifier designs.

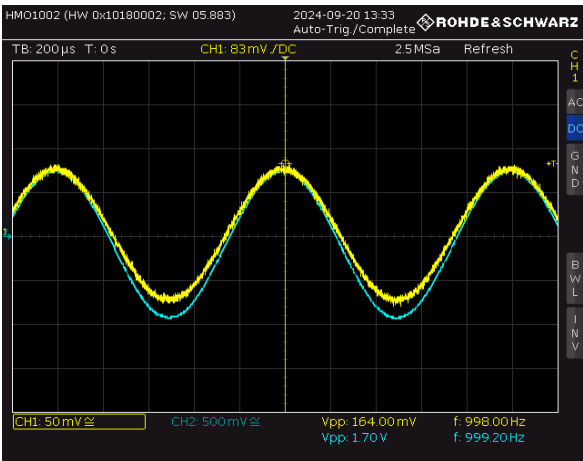
Measured resistor values using the DMM:

|               | Gain of 10               |                          | Gain of 100              |                          |
|---------------|--------------------------|--------------------------|--------------------------|--------------------------|
| Configuration | Resistor 1 (k $\Omega$ ) | Resistor 2 (k $\Omega$ ) | Resistor 1 (k $\Omega$ ) | Resistor 2 (k $\Omega$ ) |
| Inverting     | 0.9852                   | 9.881                    | 0.9852                   | 97.853                   |
| Non-Inverting | 0.9852                   | 9.881                    | 0.9852                   | 97.853                   |

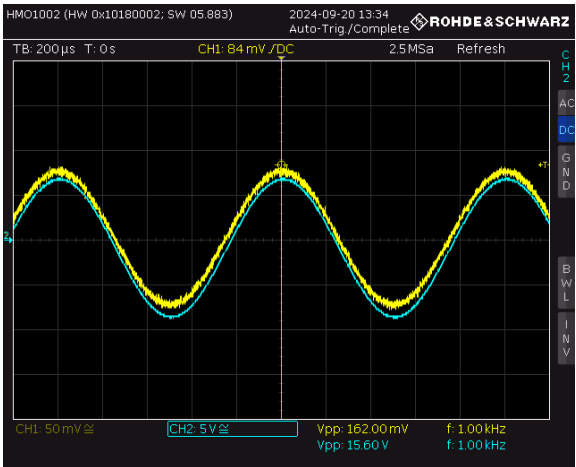
Measured voltages using the oscilloscope and calculated gain:

|               | Gain of 10    |           | Gain of 100   |           |
|---------------|---------------|-----------|---------------|-----------|
| Configuration | Non-Inverting | Inverting | Non-Inverting | Inverting |
| Vin (V)       | 0.164         | 2.10      | 0.162         | 0.154     |
| Vout (V)      | 1.70          | 21.0      | 15.60         | 14.60     |
| Gain (V/V)    | 10.37         | 10.00     | 96.30         | 94.81     |

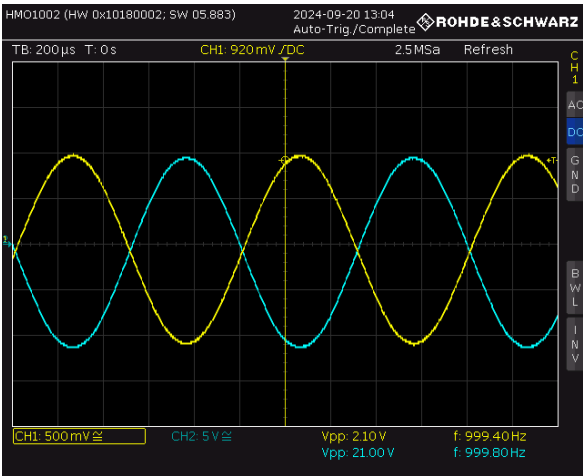
Non-inverting gain 10



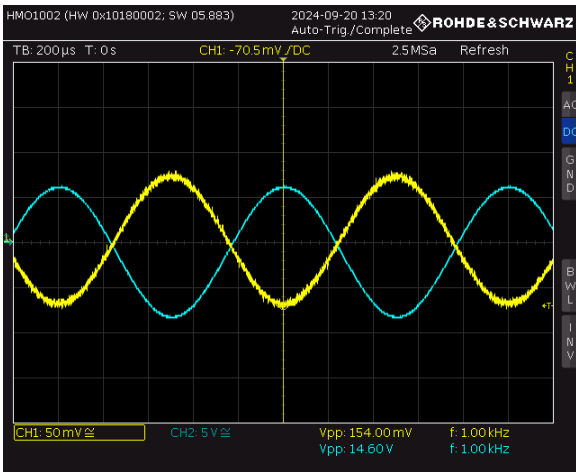
Non-inverting gain 100



Inverting gain 10

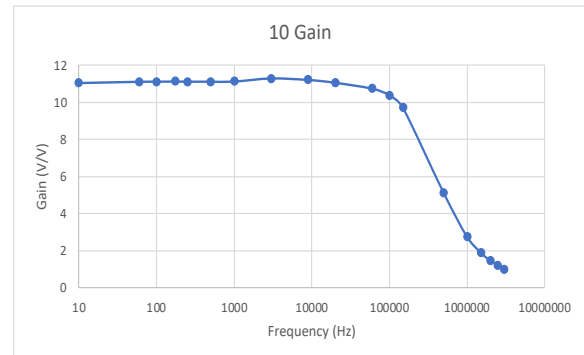
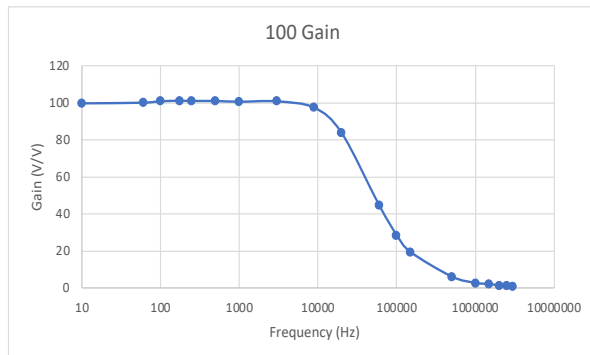


Inverting gain 100



## Experiment 2

**Introduction:** In this section, we investigate the frequency response of a non-inverting operational amplifier with gains of 100 and 10. A 1 kHz input signal is applied, and the frequency is swept from 10 Hz to 3 MHz while recording the input and output voltages. A Bode plot will be generated from the collected data, and the unity gain frequency will be determined. The experiment will analyze how the amplifier's frequency response changes with different gain settings and assess its performance at higher frequencies.



|                      |                         |                         |
|----------------------|-------------------------|-------------------------|
|                      | Experimental Value (Hz) | Experimental Value (Hz) |
|                      | Gain of 10              | Gain of 100             |
| Unity-Gain-Frequency | 3000000                 | 2500000                 |

|                             |                          |                          |
|-----------------------------|--------------------------|--------------------------|
|                             | Experimental Value (kHz) | Experimental Value (kHz) |
|                             | Gain of 10               | Gain of 100              |
| 3dB Frequency ( $f_{3dB}$ ) | 150                      | 20                       |

The frequency response experiment shows that both amplifiers maintain stable gain at low frequencies, but the gain starts to drop off significantly beyond 20 kHz. The unity gain frequency occurs around 2.5 for 100 gain and 3 MHz for 10 gain. This behavior confirms the expected bandwidth limitations of the opamp, demonstrating the trade-off between gain and frequency response, with the unity gain bandwidth being consistent with typical opamp characteristics.

### Experiment 3

#### Introduction:

Experiment 3 aims to evaluate the Offset Voltage, Bias Current, and Offset Current. To achieve this, we will employ three different circuit configurations, utilizing standard KCL and KVL analyses for op amps to derive these three DC imperfections. This experiment will illustrate how offset voltage, bias current, and offset current influence the output voltage of the op amp.

#### Calculations:

|           | Resistor 1 | Resistor 2 | DC Output |
|-----------|------------|------------|-----------|
| Circuit 1 | 0.9999MΩ   | 0.9874MΩ   | -1.7mV    |
| Circuit 2 | -          | 0.9874MΩ   | -1.62mV   |
| Circuit 3 | 0.9994KΩ   | 0.9874MΩ   | -1.72V    |

Equation #1 - Derived from Circuit 1:

$$V_o = V_{os} - I_{B_2}R_1 + I_{B_1}R_2$$
$$-1.7 \times 10^{-3} = V_{os} - I_{B_2}(0.9999 \times 10^6) + I_{B_1}(0.9874 \times 10^6)$$

Equation #2 - Derived from Circuit 2:

$$V_o = V_{os} + I_{B_1}R_2$$
$$-1.62 \times 10^{-3} = V_{os} + I_{B_1}(0.9874 \times 10^6)$$

Equation #3 - Derived from Circuit 3:

$$V_o = V_{os} \left( 1 + \frac{R_2}{R_1} \right) + I_{B_1}R_2$$
$$-1.72 = V_{os} \left( 1 + \frac{0.9874 \times 10^6}{0.9994 \times 10^3} \right) + I_{B_1}(0.9874 \times 10^6)$$

Solving systems of equations using Matlab:

Input:

```
format long
A = [1, -0.9999*10^6, 0.9874*10^6, -1.7*10^-3;
     1, 0, 0.9874*10^6, -1.62*10^-3;
     (1 + ((0.9874*10^6)/(0.9838*10^3))), 0, 0.9874*10^6, -1.718];
rref(A)
```

Output:

```
-0.001710122183512 = Vos
0.000000000080008 = IB1
0.000000000091272 = IB1
```

Calculating bias and offset current using obtained  $I_{B1}$  and  $I_{B2}$  values:

$$\text{Bias Current: } I_B = \frac{(I_{B1} + I_{B2})}{2} = \frac{((80.01 \times 10^{-12} \text{c}) + (91.27 \times 10^{-12}))}{2} = 85.64 \text{pA}$$

$$\text{Offset Current: } I_{os} = |I_{B1} - I_{B2}| = |((80.01 \times 10^{-12}) - (91.27 \times 10^{-12}))| = 11.26 \text{pA}$$

|                | Calculated Values |
|----------------|-------------------|
| Offset Voltage | -1.71mV           |
| Bias Current   | 85.64pA           |
| Offset Current | 11.26pA           |

### Comparison:

When comparing our calculations with the TL084 data sheet values, here is what we observe:

|                | Data Sheet Minimum  | Data Sheet Maximum  | Calculated Values |
|----------------|---------------------|---------------------|-------------------|
| Offset Voltage | $\pm 1 \text{mV}$   | $\pm 4 \text{mV}$   | -1.71mV           |
| Bias Current   | $\pm 1 \text{pA}$   | $\pm 120 \text{pA}$ | 85.64pA           |
| Offset Current | $\pm 0.5 \text{pA}$ | $\pm 120 \text{pA}$ | 11.26pA           |

When we compared our experimental results to the values in the TL084 data sheet, we found some interesting insights. Our calculated offset voltage of -1.71 mV is within the specified range, but it's a bit more negative than the minimum of  $\pm 1 \text{ mV}$ . This shows that while our measurement is close, there's some variability you might expect with real components. For the bias current, we measured 85.64 pA, which is close to the upper limit of  $\pm 120 \text{ pA}$  in the datasheet. This suggests that the op-amp we tested might have slightly higher bias currents than average. Finally, our offset current of 11.26 pA falls nicely within the range of  $\pm 0.5 \text{ pA}$  to  $\pm 120 \text{ pA}$ , indicating that it performs well. Overall, our results highlight how important it is to consider variations in real-world components and how our experimental setup did a good job capturing these parameters accurately.

## Experiment 4

### Introduction:

In this experiment, we explore the functionality of a Schmitt Trigger, a comparator circuit known for its hysteresis characteristics. By selecting and measuring specific resistor values, we will calculate the expected toggle (trigger) voltages. We aim to compare the theoretical calculations

with actual measurements and LT-Spice simulations, enhancing our understanding of the circuit's behavior.

Resistor Values:

| Resistor 1 | Resistor 2 | Resistor A |
|------------|------------|------------|
| 14.878kΩ   | 1.773kΩ    | 1.7805kΩ   |

Theoretical calculations for trigger voltages:

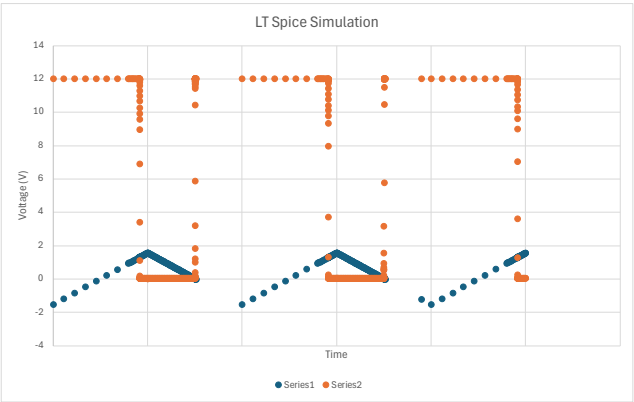
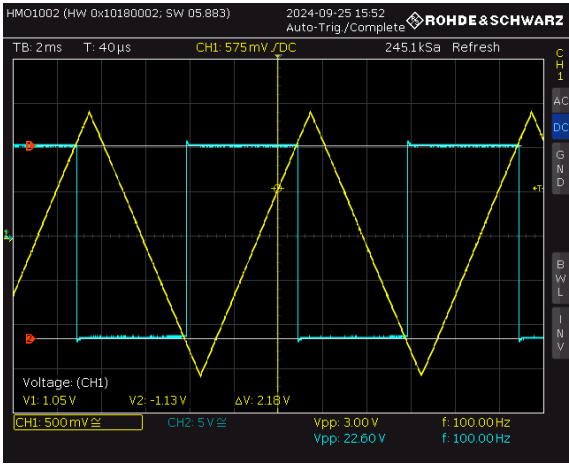
Upper Limit Trigger Value:

$$V_{UTP} = \frac{R_2}{(R_1 + R_2)} V_{out} (Max)$$
$$V_{UTP} = \frac{(1.773 \times 10^3)}{(14.878 \times 10^3) + (1.773 \times 10^3)} (12) = 1.278V$$

Lower Limit Trigger Value:

$$V_{UTP} = \frac{R_2}{(R_1 + R_2)} V_{out} (Min)$$
$$V_{UTP} = \frac{(1.773 \times 10^3)}{(14.878 \times 10^3) + (1.773 \times 10^3)} (-12) = -1.278V$$

Experimental Values (Measured from Oscilloscope) and LT Spice Simulation:



|              | $V_{UTP}$ | $V_{LTP}$ | $V_{out} (Max)$ | $V_{out} (Min)$ |
|--------------|-----------|-----------|-----------------|-----------------|
| Calculated   | 1.278V    | -1.278V   | 12V             | -12V            |
| Experimental | 1.05V     | -1.13V    | 12V             | -12V            |
| LT-Spice     | 1.258     | -1.258    | 12V             | -12V            |

## Comparison:

The comparison of toggle voltage values from theoretical calculations, experimental measurements, and LT-Spice simulations reveals both alignment and discrepancies. The calculated upper toggle point ( $V_{UTP}$ ) was 1.278V, while the experimental value was 1.05V, and the LT-Spice result was 1.258V. For the lower toggle point ( $V_{LTP}$ ), the calculated value of -1.278V compared to -1.13V experimentally and -1.258V in simulation. Despite these variances, likely due to component tolerances, the output voltages consistently measured 12V (max) and -12V (min) across all methods. These results demonstrate the effectiveness of theoretical predictions and simulations in approximating circuit behavior, emphasizing the value of empirical validation.

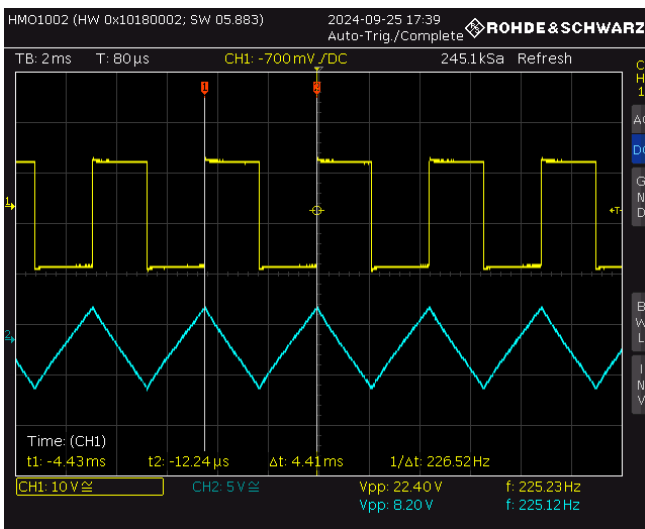
## Experiment 5

**Introduction:** This experiment required us to construct, power and analyze the circuit shown in schematic-3 to obtain wave forms without using a Function Generator. Once the circuit was constructed and powered using only the power supply the input and output probes were attached to the nodes A, B, and C, respectively and resulted in waveforms appearing on the oscilloscope.

## Measurements:

| <i>Schematic</i> | $R_1(5.6k\Omega)$ | $R_2(2.2k\Omega)$ | $R_3(22k\Omega)$ | $C(0.1\mu F)$  | $V_{EE}$ | $V_{CC}$ |
|------------------|-------------------|-------------------|------------------|----------------|----------|----------|
| <i>Measured</i>  | 5.546k $\Omega$   | 2.157k $\Omega$   | 21.873k $\Omega$ | 0.1015 $\mu F$ | -12.031  | 12.042   |

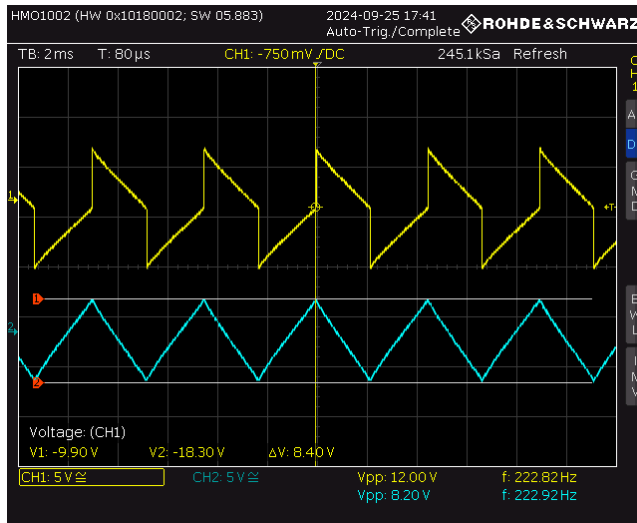
## Results:



Oscilloscope with period value displayed

Channel 1, A: Yellow

Channel 2, B: Blue



Oscilloscope with period value displayed

Channel 1, C: Yellow

Channel 2, B: Blue

### Period Calculation:

$$Frequency = \frac{1}{4R_2C} \left( \frac{R_1}{R_3} \right) = \frac{1}{4(2.157 \times 10^3)(0.1015 \times 10^{-6})} \left( \frac{5.546 \times 10^3}{21.873 \times 10^3} \right) = 289.531$$

$$Period = \frac{1}{f} = \frac{1}{289.531} = 0.00345386 = 3.45ms$$

| Node A            | Node B                                                                                                               | Node C                                                                                                                                                                                                                                               |
|-------------------|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $V_{AMAX} = 12V$  | $V_{BMAX} = V_{UTP}$<br>$= \frac{R_2}{R_1} * V_{MAX}$<br>$= \frac{2.157 * 10^3}{5.546 * 10^3} (12)$<br>$= 4.66715$   | $V_{CMAX} = V_{AMAX} \frac{R_2}{R_1 + R_2} + \left( \frac{V_{BMAX} R_1}{R_1 + R_2} \right)$<br>$= (12) \frac{2.157 * 10^3}{5.546 * 10^3 + 2.157 * 10^3} \left( \frac{(4.66715) * 5.546 * 10^3}{5.546 * 10^3 + 2.157 * 10^3} \right)$<br>$= 11.291$   |
| $V_{AMIN} = -12V$ | $V_{BMIN} = V_{LTP}$<br>$= \frac{R_2}{R_1} * V_{MIN}$<br>$= \frac{2.157 * 10^3}{5.546 * 10^3} (-12)$<br>$= -4.66715$ | $V_{CMIN} = V_{AMIN} \frac{R_2}{R_1 + R_2} + \left( \frac{V_{BMIN} R_1}{R_1 + R_2} \right)$<br>$= (-12) \frac{2.157 * 10^3}{5.546 * 10^3 + 2.157 * 10^3} \left( \frac{(4.66715) * 5.546 * 10^3}{5.546 * 10^3 + 2.157 * 10^3} \right)$<br>$= -11.291$ |

### Comparison:

In comparing the calculated, experimental, and simulated values for the Schmitt Trigger, we found that they aligned quite well, especially for the toggle points. The calculated upper toggle point ( $V_{BMAX}$ ) was about 4.667 V, which matched closely with what we measured experimentally. The lower toggle point ( $V_{BMIN}$ ) was also consistent at -4.667 V. For Node C,

the maximum and minimum voltages confirmed that the circuit was behaving as expected across all methods. These results really validate the theoretical models and simulations we used in the design, showing that the circuit works effectively, even with some small differences due to component tolerances. Overall, this experiment highlights how important it is to combine theory with real-world testing to ensure reliable circuit performance.

### **Conclusion:**

In conclusion, this lab provided valuable insights into the practical applications and characteristics of operational amplifiers. Through the design and implementation of non-inverting and inverting amplifiers, we observed how resistor values directly influence the gain and confirmed the theoretical values through experimental results. The frequency response tests revealed the bandwidth limitations of opamps and the unity gain frequency. Additionally, by measuring offset voltage, bias current, and offset current, we characterized the imperfections inherent in real-world opamps, comparing them to the datasheet specifications. The Schmitt Trigger circuit successfully demonstrated hysteresis, and the free-running oscillator generated predictable waveforms, which were verified through both experimental observations and LT-Spice simulations. This lab reinforced the importance of precision in component measurements and highlighted the differences between ideal and real-world circuit behavior, enhancing our understanding of opamp-based circuits in practical applications.